Observation

Mass of the circular plate M = 0.645

Radius of circular plate R=(Circumference/ 2π)= 0.05

Mass of the rectangular plate M = 0.66

Length of the rectangular plate L = 0.12

Breadth of the rectangular plate B = 0.08

Length of the wire between the chuck nuts l= 0.25

Radius of the given wire r = 0.3*10^-3

Body	Axis	Moment of Inertia (I) Kg-m2	Time taken for 10 oscillations 't' sec			Period T = t /10 sec	T^2	I / T^ 2 kg-m/s^2
			t1	t2	(t1+t2) /2			
Rectangular plate	Perpendicular to its plane	0.00114	21.000	22.000	21.500	2.150	4.623	0.000247
	Perpendicular to its length	0.00079	19.000	19.000	19.000	1.900	3.610	0.000219
	Perpendicular to its breadth	0.00035	12.000	14.000	13.000	1.300	1.690	0.000208
Circular plate	Perpendicular to its plane	0.00081	24.000	23.000	23.500	2.350	5.523	0.000146
	Along the diameter	0.00040	15.000	15.000	15.000	1.500	2.250	0.000179
Irregular plate	Perpendicular to its plane		20.000	20.000	20.000	2.000	4.000	

Mean I/t^2 0.0002

Calculation

(1) Moment of Inertia of the irregular body (Perpendicular to the plane)

I =T^2*(I/T^2) 0.0008 Kgm^2

(2) Rigidity modulus of the given material wire

η 1.55139E+11 Nm^-2